ODSC AI

The #1 AI Builders Conference

AGENTIC AI & AUTONOMOUS SYSTEMS

SAN FRANCISCO | OCT 27-29

GO ON A SCAVENGER HUNT WITH A GEO-ENABLED, ENTITY-RESOLVED KNOWLEDGE GRAPH



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Go On a Scavenger Hunt with a Geo-Enabled Entity-Resolved Knowledge Graph

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& ASSOCIATES

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github.com/cj2001/odsc_west_2026

Run SETUP.md

Outline

- Terminology: graphs and knowledge and entities, oh my!
- Why is entity resolution (ER) hard
- Why does ER matter for RAG?
- An introduction to geocoding: the importance of “where”
- Coding and the scavenger hunt!
- Why not just use an LLM for ER?
- Conclusion

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Some definitions

- Graph
- Knowledge graph (KG)
- Entity resolution (ER)
- Entity-resolved knowledge graph (ERKG)
- Geo-coding
- The geo-ERKG

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What is an entity?

- Distinct thing
  - Person
  - Place
  - Organization
  - Object
- Think of them like rows in a table
- Defined by a combination of distinguishing characteristics
  - Some characteristics are more distinguishing than others!

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Consider the following scenarios...

Five different people, each with a single transaction

A single person with five different transactions

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Why does it matter?

- Counting records is the most basic task of a data scientist
- If you can’t count, you can’t predict
- If you can’t count, you can’t quantify
- Enables things like:
  - Good news / bad news
  - Red light / green light
  - To market to them or not
  - Benign / threat

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Hot take: regex is not the answer

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Names are complicated

Johnathan Edward Smith	Jonathan Edward Smith	Jonathon Edward Smith	Jonothan Edward Smith	Jonatan Edward Smith	John Edward Smith	Jon Edward Smith	Johnny Edward Smith
Jonny Edward Smith	Jack Edward Smith	Jono Edward Smith	Nate Edward Smith	Johnathan Ed Smith	Jonathan Ed Smith	Jonathon Ed Smith	Johnathan Eddie Smith
Jonathan Eddie Smith	Jon Eddie Smith	Johnathan Ted Smith	Johnny Ted Smith	Johnathan Teddy Smith	Johnathan Ned Smith	Johnathan E. Smith	Jonathan E. Smith
Jonathon E. Smith	Jon E. Smith	John E. Smith	Jack E. Smith	Jono E. Smith	Johnny E. Smith	J. Edward Smith	J. E. Smith
J.E. Smith	J. Smith	Jon Smith	John Smith	Johnny Smith	Jack Smith	Jono Smith	Eddie Smith
Ted Smith	Teddy Smith	Ned Smith	Nate Smith	Johnathan Edward Smithe	Jonathan Edward Smithe	Jonathon Edward Smithe	Jon E. Smithe
Johnny E. Smithe	Johnathan E. Smyth	Jonathan E. Smyth	Jon Smyth	Johnny Smyth	John Smyth	Jack Smyth	Jono Smyth
J. Edward Smyth	J. E. Smyth	J.E. Smyth	Johnathan Edward Smythe	Jonathan Edward Smythe	Jonathon E. Smythe	Jon Smythe	John Smythe
Johnny Smythe	J. E. Smythe	Ted Smythe	Eddie Smythe	Nate Smythe			

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Addresses are REALLY complicated

123 Main, Apt 1A, Las Vegas, NV, 89111	123 Main, #1A, Las Vegas, Nevada, 89111	123 Main St, Apt 1A, Las Vegas, NV, 89111	123 Main St, #1A, Las Vegas, Nevada, 89111	123 Main St., Apt 1A, Las Vegas, NV, 89111	123 Main St., #1A, Las Vegas, Nevada, 89111	123 Main Street, Apt 1A, Las Vegas, NV, 89111	123 Main Street, #1A, Las Vegas, Nevada, 89111
123 Main, Apt 1A, Las Vegas, Nevada, 89111	123 Main, Apartment 1A, Las Vegas, NV, 89111	123 Main St, Apt 1A, Las Vegas, Nevada, 89111	123 Main St, Apartment 1A, Las Vegas, NV, 89111	123 Main St., Apt 1A, Las Vegas, Nevada, 89111	123 Main St., Apartment 1A, Las Vegas, NV, 89111	123 Main Street, Apt 1A, Las Vegas, Nevada, 89111	123 Main Street, Apartment 1A, Las Vegas, NV, 89111
123 Main, Apt. 1A, Las Vegas, NV, 89111	123 Main, Apartment 1A, Las Vegas, Nevada, 89111	123 Main St, Apt. 1A, Las Vegas, NV, 89111	123 Main St, Apartment 1A, Las Vegas, Nevada, 89111	123 Main St., Apt. 1A, Las Vegas, NV, 89111	123 Main St., Apartment 1A, Las Vegas, Nevada, 89111	123 Main Street, Apt. 1A, Las Vegas, NV, 89111	123 Main Street, Apartment 1A, Las Vegas, Nevada, 89111
123 Main, Apt. 1A, Las Vegas, Nevada, 89111	123 Main, Unit 1A, Las Vegas, NV, 89111	123 Main St, Apt. 1A, Las Vegas, Nevada, 89111	123 Main St, Unit 1A, Las Vegas, NV, 89111	123 Main St., Apt. 1A, Las Vegas, Nevada, 89111	123 Main St., Unit 1A, Las Vegas, NV, 89111	123 Main Street, Apt. 1A, Las Vegas, Nevada, 89111	123 Main Street, Unit 1A, Las Vegas, NV, 89111
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Different character sets

جوناثان إدوارد سميث

A شارع مين، شقة 1 123

لاس فيغاس، نيفادا 89111

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Cultural variations

Argentina: Piedras 623, Piso 2, depto 4, C1070AAM, Capital Federal

Belgium: Rue du Vivier 7C bte 5, 1000 Bruxelles, BELGIQUE

India: e-506 street number 78, uttam viharblock d, uttam nagar, bindapur, 110059, new delhi

New Zealand: 43 Vogel Street, Roslyn, Palmerston North 4414

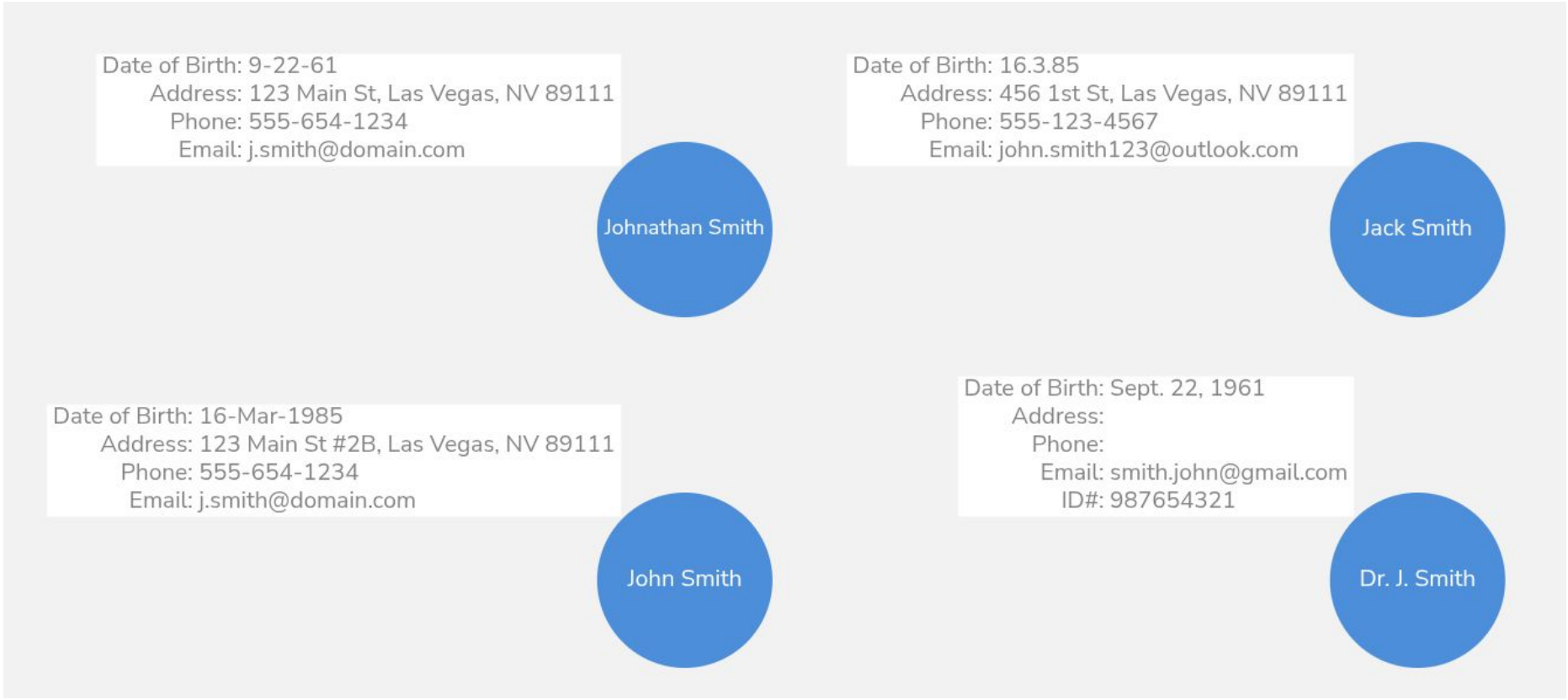
Ukraine: vul. Shevchenka, bud. 17, m. Bila Tserkva, Kyivs'ka obl., 09117, UKRAINE

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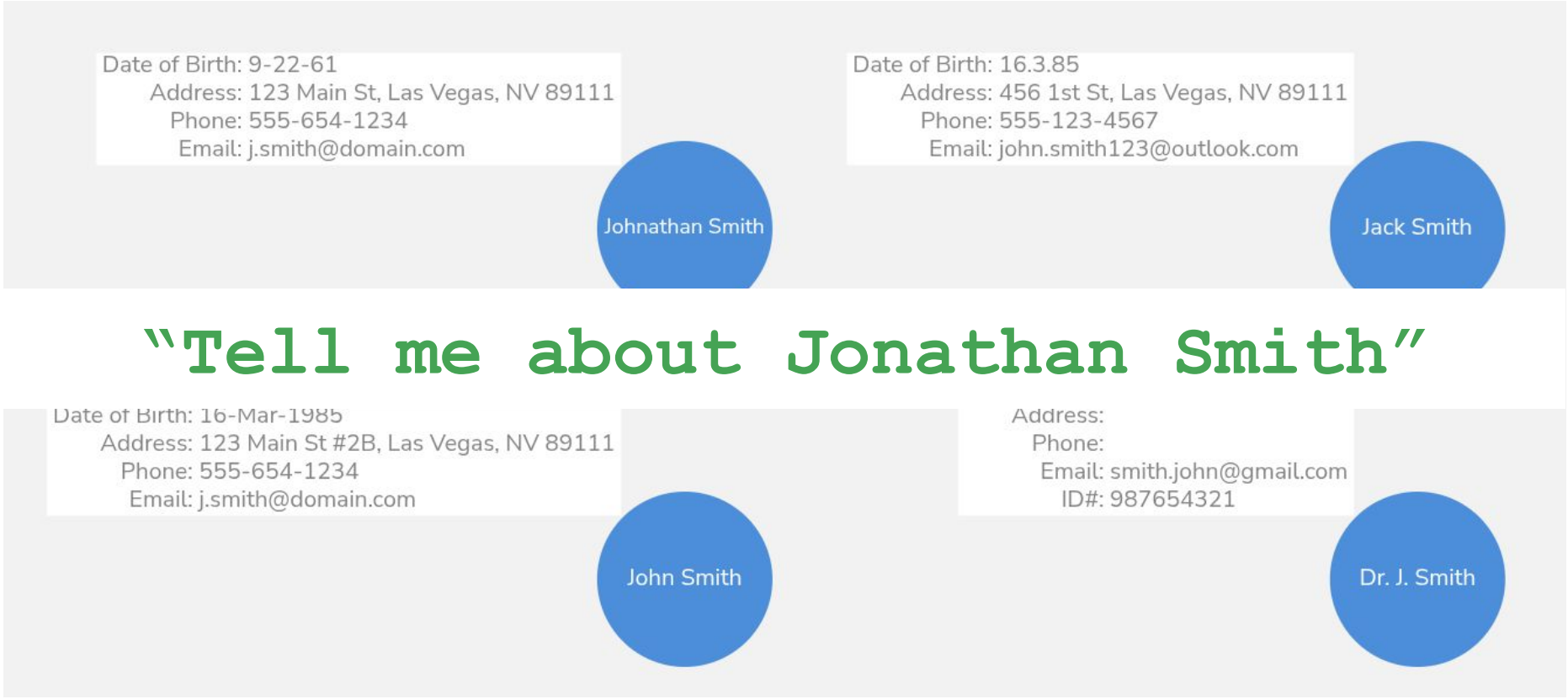
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Your RAG is only as good as your ER!

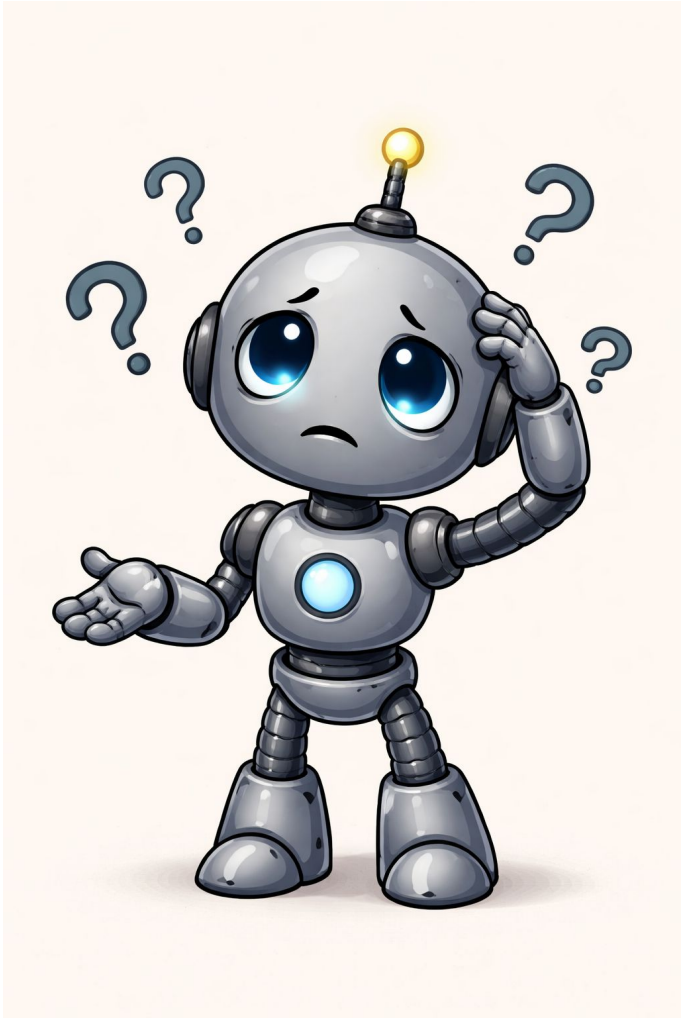
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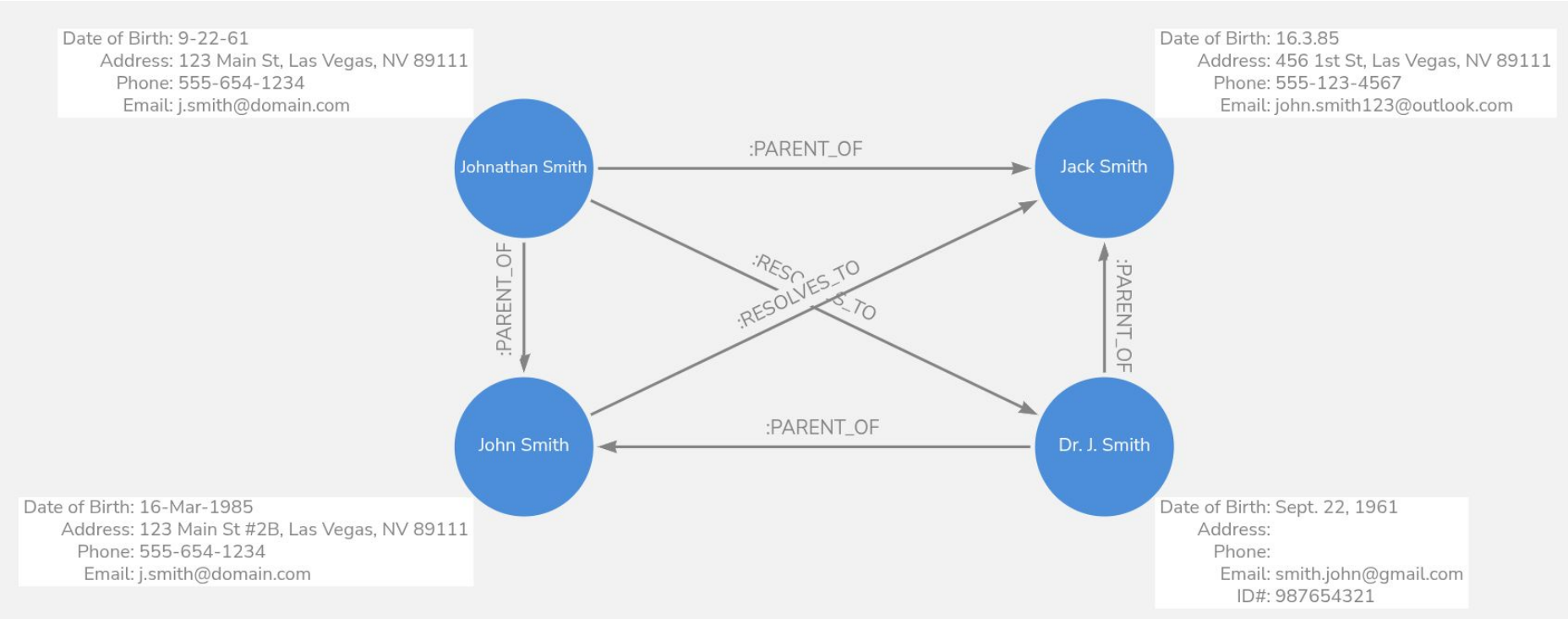


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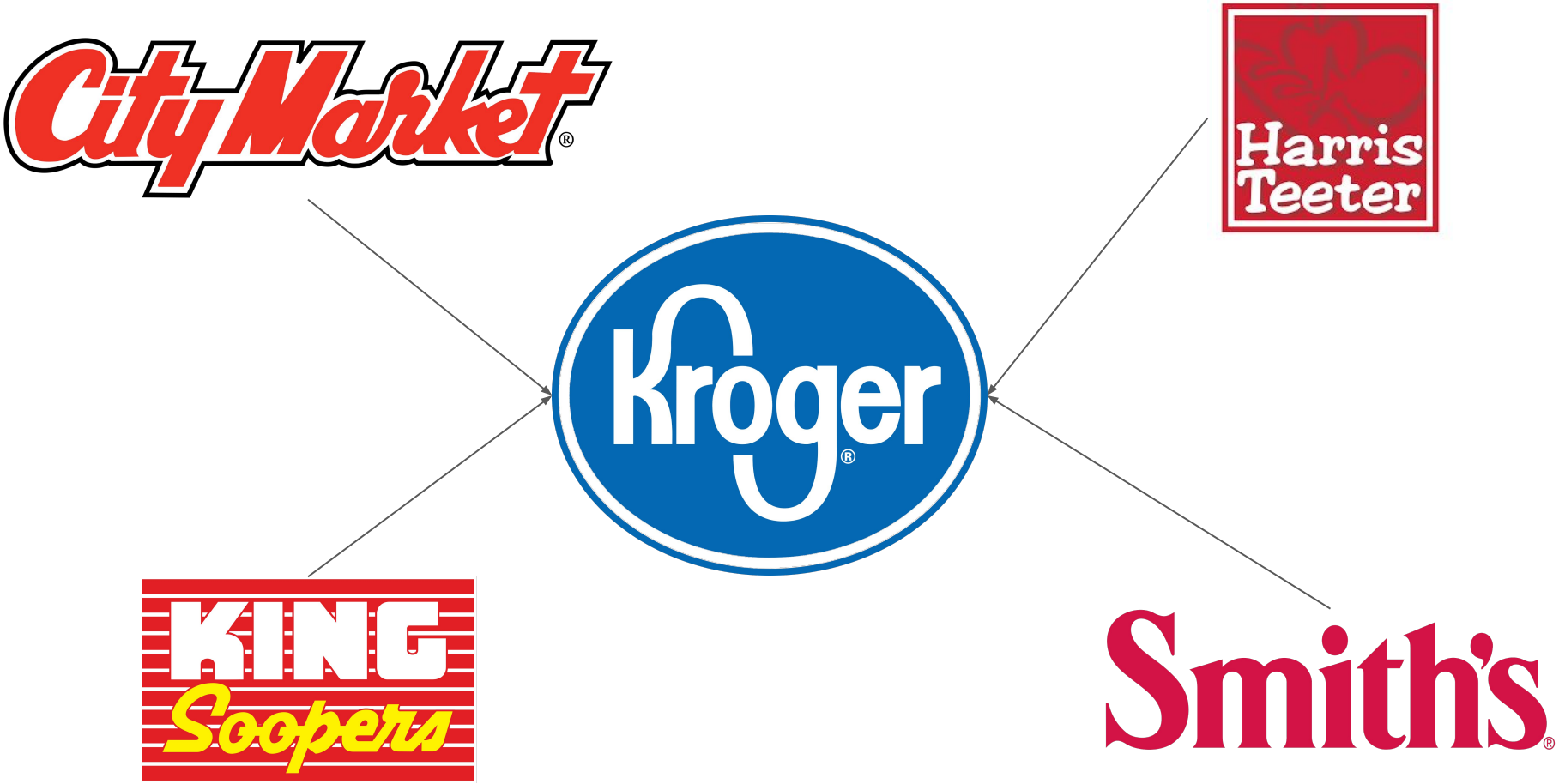


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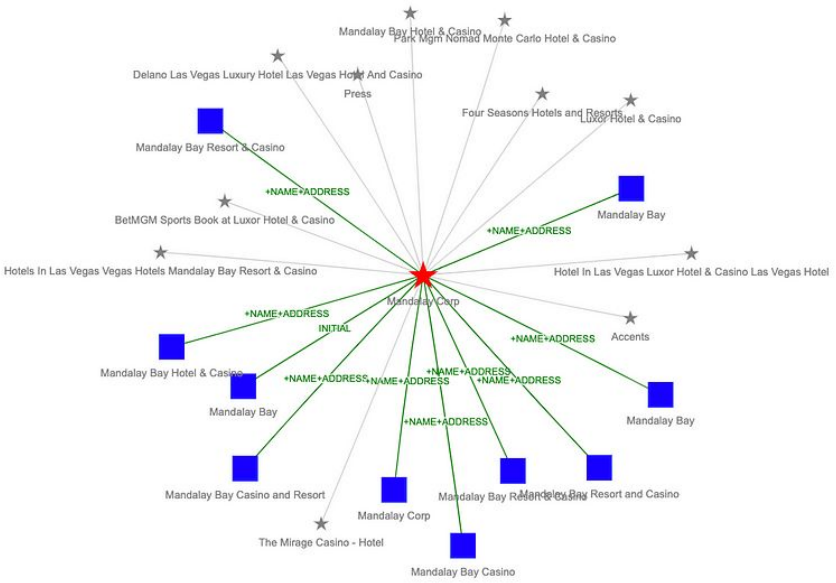
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ER is naturally a graph problem: the ERKG

- **Graphs provide more context to your RAG problems, resulting in better answers than vector-based solutions!**
- Records resolve into entities
- The relationships between the nodes can represent how those entities resolved (MATCH KEY)
  - Name
  - Address
  - Phone
  - Registration or identification numbers
  - Etc.
- ER can surface disclosed and undisclosed connections, possible relationships



https://neo4j.com/blog/developer/entity-resolved-knowledge-graphs/

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The importance of “where” – from records to places

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What is geocoding?

- Turning a description of a place into a (string) position
- Address - coordinate - string
- But why not just use GPS coordinates???
  - An address is a string
  - Strings answer exactly one question: are these two identical?
  - And they are bad at that!
    - Ex: 1494 BAYSHORE HWY vs. 1494 Bayshore Highway vs. 1494 Bayshore Hwy
- JOINing on GPS coordinates is a really bad idea!
  - Ex: (37.5921, -122.3598) vs. (37.5922, -122.3597)
  - 11 m apart, almost certainly the same storefront
  - ...but you can't join on a measurement so you have to rely on distance thresholding
- A string allows for checking equality

Equality is free. Proximity/distance is computation.

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What is geocoding? (cont.)

- Goal
  - Stable: same place in, same value out
  - JOINable: equality, not distance
  - How precise are GPS coordinates really?
    - 6 decimate points = cm accuracy...but really???
    - Coordinates present an estimate as a fact
  - Decide whether two things are at the same place
    - Coordinates are great for determining distance, but not at answering this
  - A **key** we can join on

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Introduction to H3

- An open source discrete global grid system for indexing geographies into a hexagonal grid, developed at Uber
- Hexagons because every neighbor is the same distance away
- Hierarchical geospatial index
- 16 resolutions
  - Nested
  - Resolution 0: Continent
  - Resolution 15: 0.9 m^2/cell (edge length: ~0.5 m)



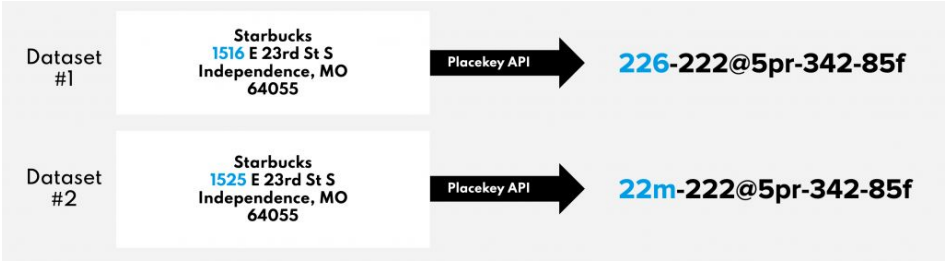
https://h3geo.org/docs/highlights/indexing

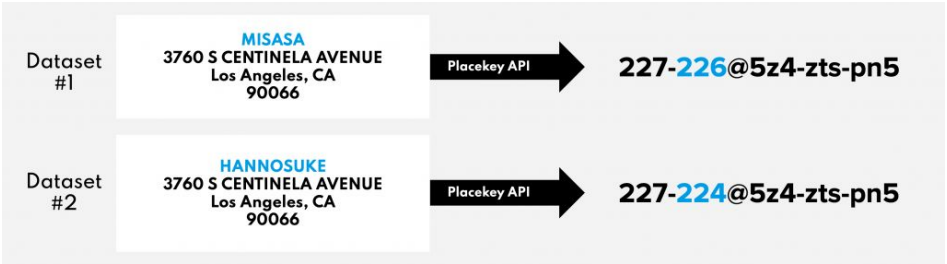
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Placekey

- An open identifier for any physical place
- What @ Where
  - Where = H3 Resolution 10 cell
    - Hexagon side ~ 122 m
  - What = address and point of interest (POI) encoding
- 3 potential JOIN keys
  - Same full key = same business
  - Same what = same address
  - Same where = same block/region
- Ex: Starbucks inside Target
  - One address, two POIs
- Note: Placekey only gets a POI encoding if Placekey's index already knows about that business. Otherwise it just does address encoding.







https://digital-geography.com/joining-data-with-the-placekey-qgis-plugin/

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What can geocoding add to our ERKG?

- Augments ERKG in a meaningful way
  - ER reads names and addresses
  - Geocoding reads position
- Adds proximity relationship to graph
- Location opens up new questions that can be asked
- ER and geocoding can complement each other on data quality tasks
- Three granularities with one key
- An answer we can walk to!

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Run SETUP.md

The code

- 00: Verifies workshop environment is properly configured and running
- 01: Conduct ER on a known dataset (OPTIONAL)
- 02: EDA of workshop database
- 03: It's a graph. It's geo. It's a GEO-GRAPH!!!
- 04: Traditional RAG approach (no ER, no graphs, no geo)
- 05: Full geo-ERKG RAG

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The data

- All publicly available and downloadable
  - Contained in the container database
  - Also available in CSV format from GitHub repository
- CA Secretary of State business registrations
- Burlingame Chamber of Commerce business listings
- Master list of CA-licensed contractor businesses and personnel
- CA Alcoholic Beverage Control (ABC) listings
  - Retail and non-retail

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Let’s code!

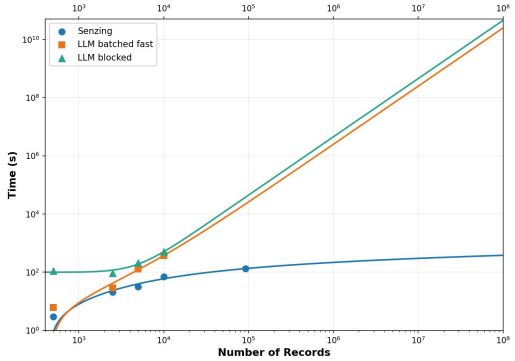
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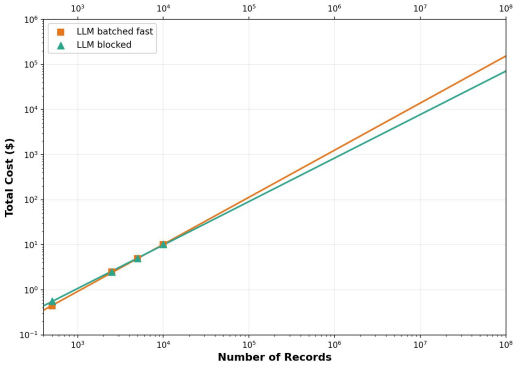
What if we just did ER with GenAI?

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Comparing an ER engine to an LLM

- Took identical datasets of different sizes and did ER with both Senzing and Claude
- **Purpose-built ER engine was *significantly* faster and cheaper than allowing an LLM to do the same problem**
- Purpose-built ER engine scaled linearly with the number of records
- The LLM must be re-run every time new data arrives!
- The LLM was wrong a significant amount of the time
  - Both false positives and false negatives
- The only way to improve this is to completely rebuild ER within the LLM, which defeats the purpose





Conclusions

- Your RAG is only as trustworthy as the data underneath it
- If you cannot accurately identify who or what your entities are, you cannot count them, relate them, or reason about them
- A knowledge graph built on unresolved entities is actively misleading
- Adding location with geocodes to your ERKG opens up many questions that can be asked of your RAG
- geoERKGs give your LLM something vector search, RAG, and traditional GraphRAG can not: **REAL WORLD context**

pollEv.com/clairsullivan399



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Thank you!

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